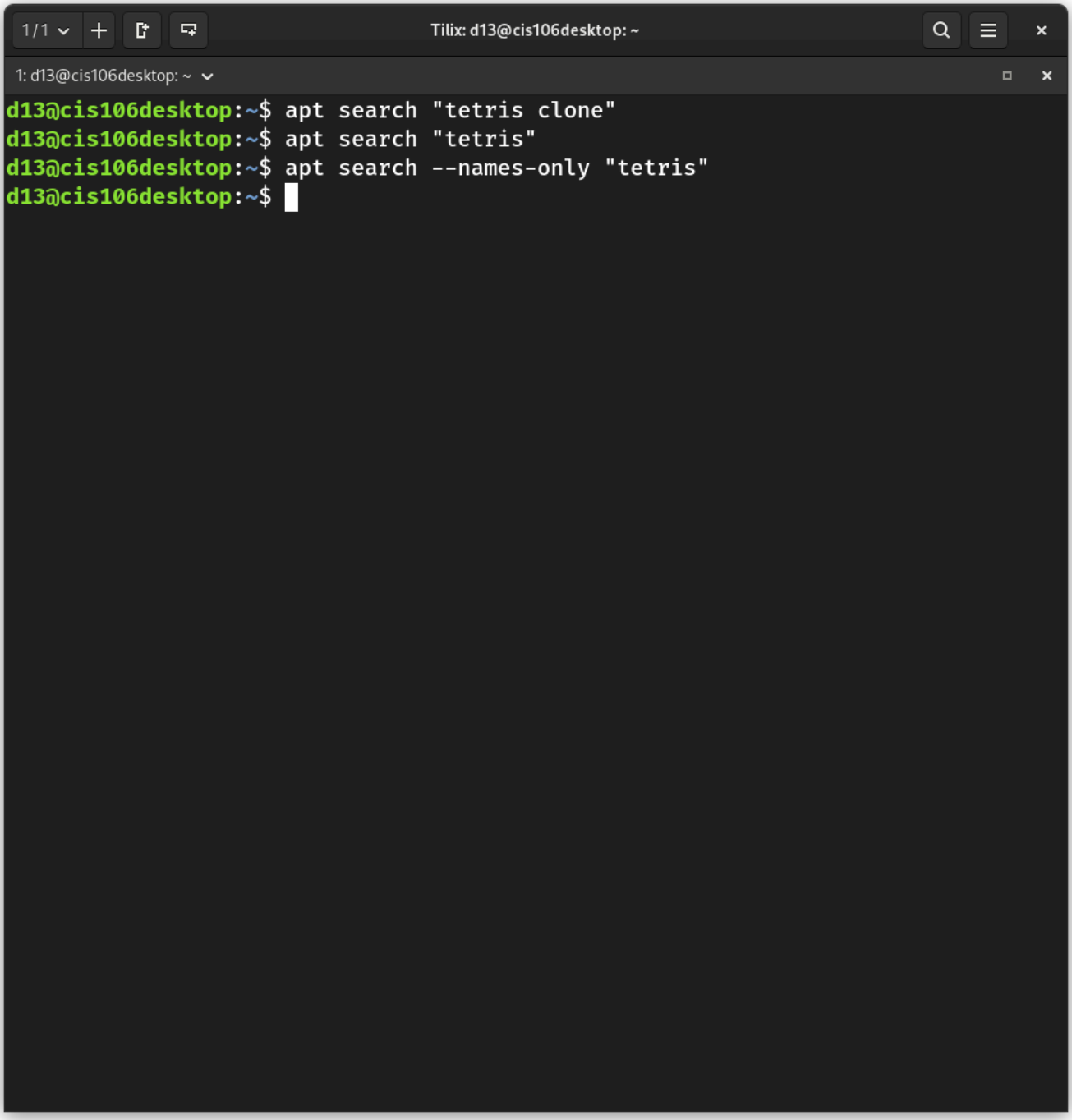
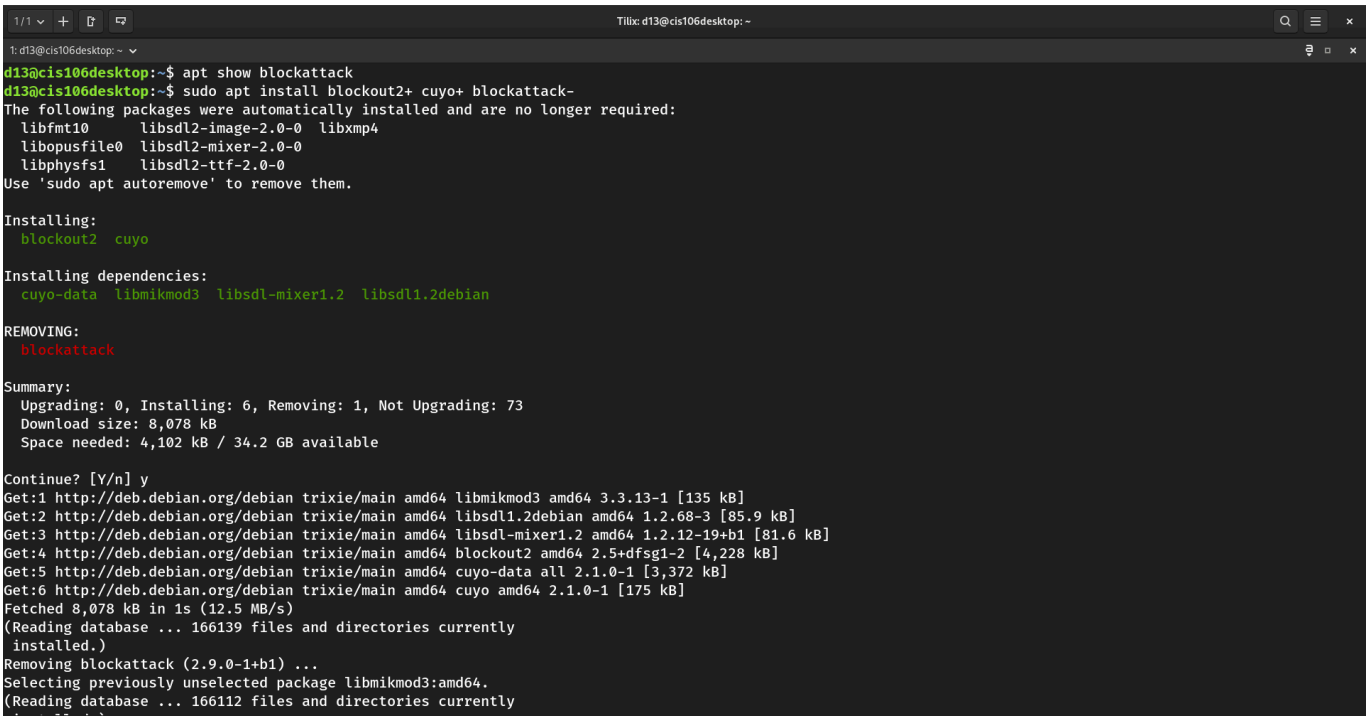
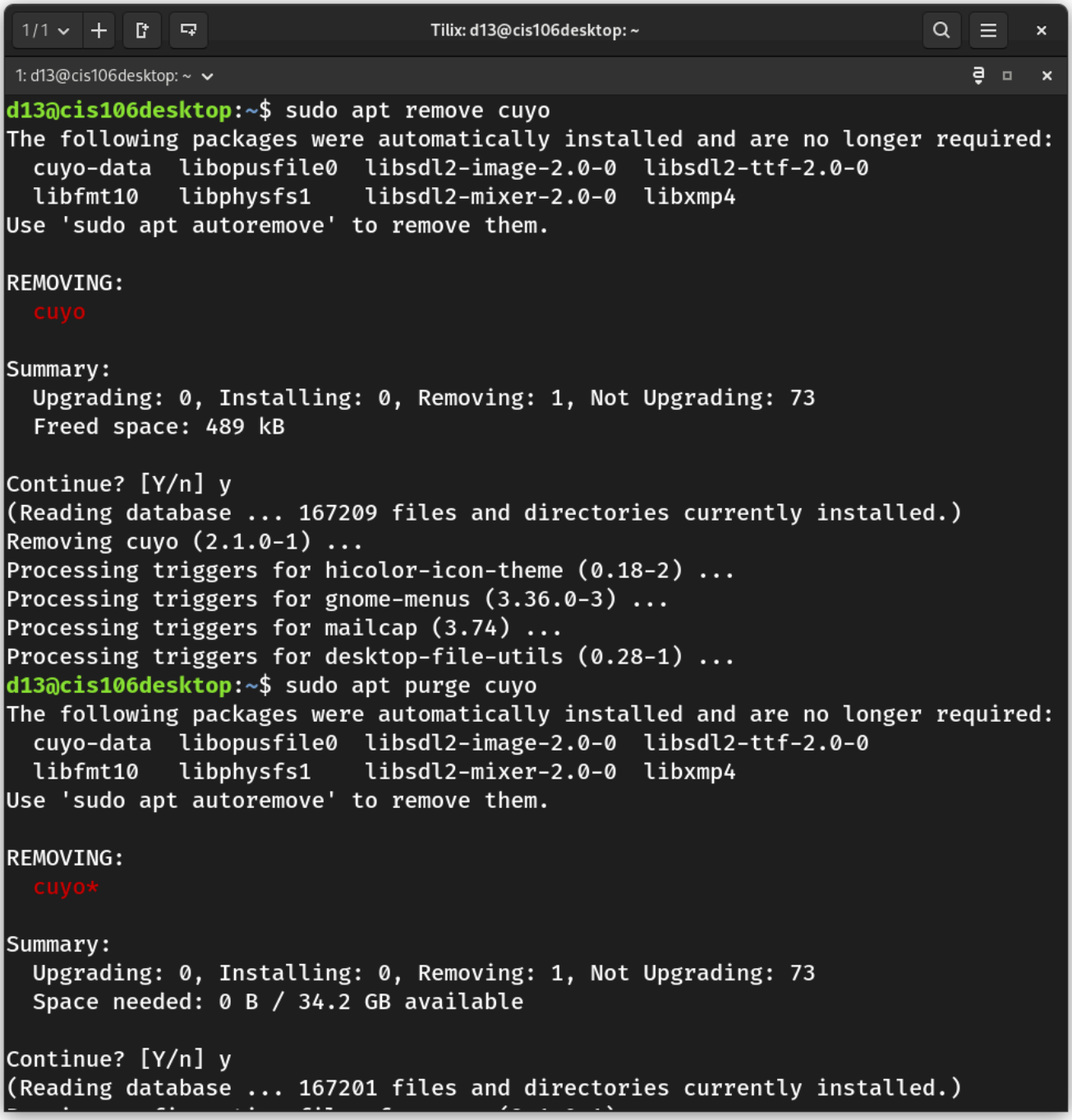
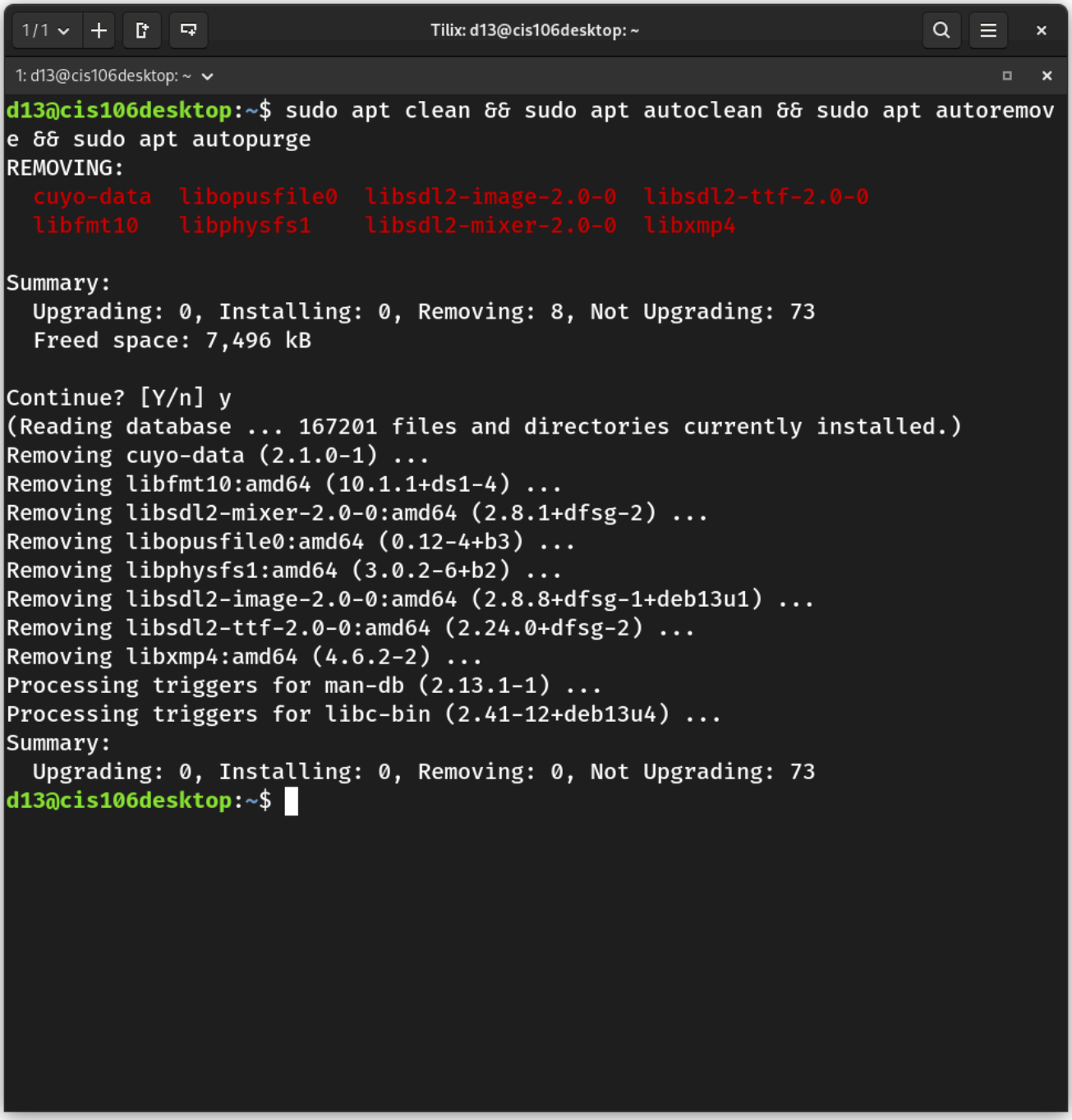
Lab 4 Submission

Question 1

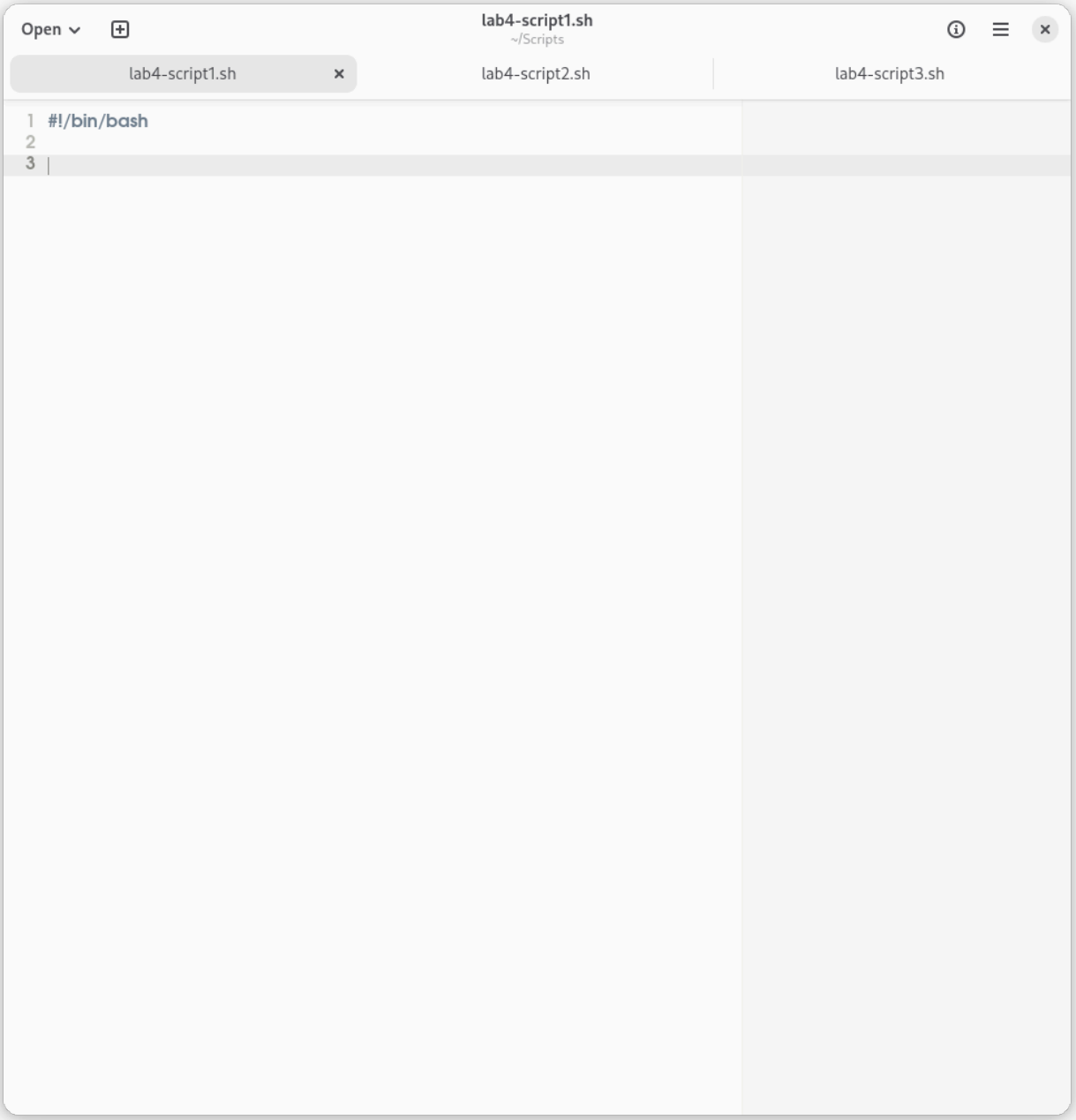

1 / 1
Tilix: d13@cis106desktop: ~
1: d13@cis106desktop: ~
d13@cis106desktop:~$ apt search "tetris clone"
d13@cis106desktop:~$ apt search "tetris"
d13@cis106desktop:~$ apt search --names-only "tetris"
d13@cis106desktop:~$

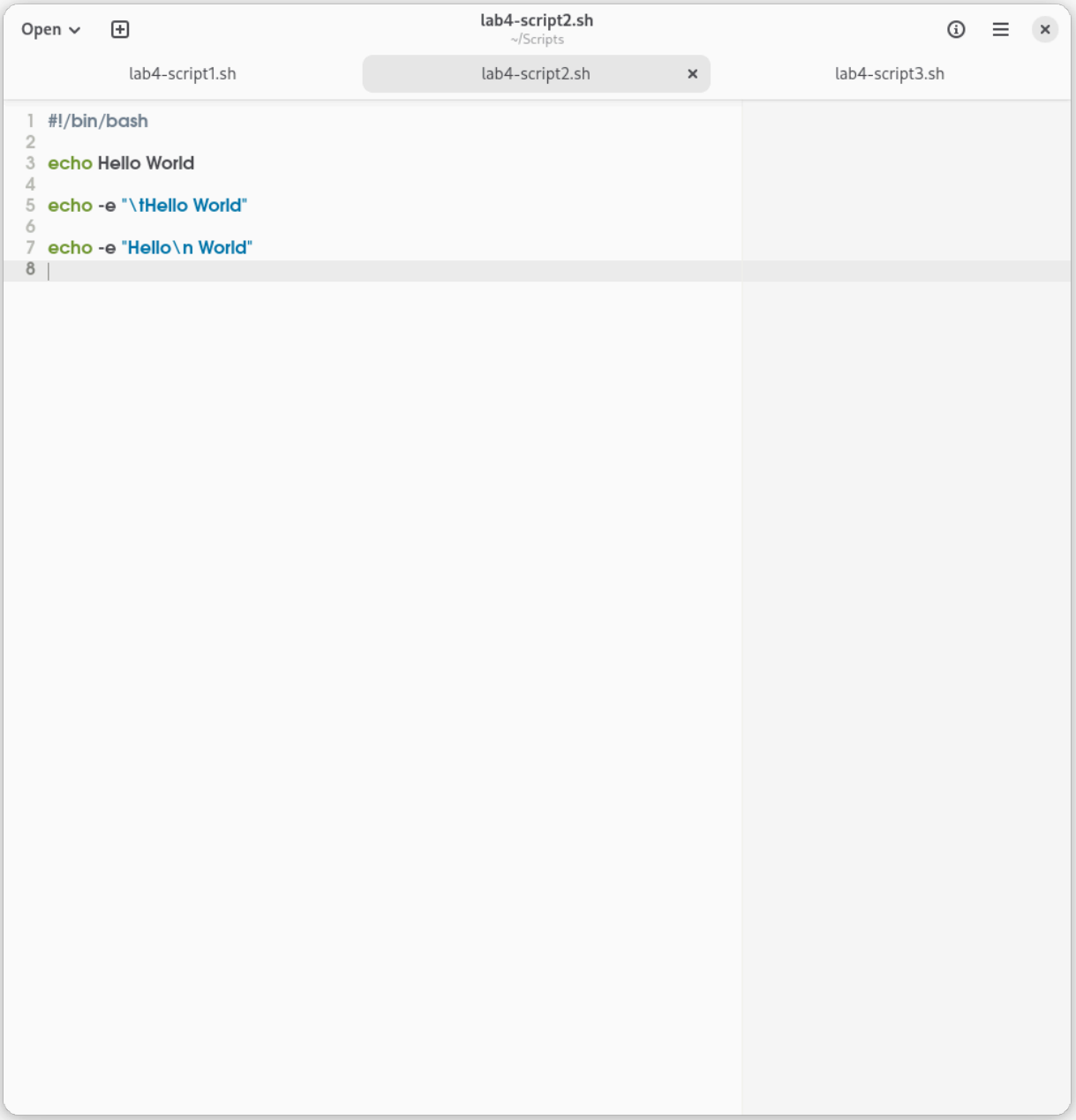

1 / 1
Tilix: d13@cis106desktop: ~
1: d13@cis106desktop: ~
d13@cis106desktop:~$ apt show blockattack
d13@cis106desktop:~$ sudo apt install blockout2+ cuyo+ blockattack-
The following packages were automatically installed and are no longer required:
libfmt10 libsdl2-image-2.0-0 libxmp4
libopusfile0 libsdl2-mixer-2.0-0
libphysfs1 libsdl2-ttf-2.0-0
Use 'sudo apt autoremove' to remove them.
Installing:
blockout2 cuyo
Installing dependencies:
cuyo-data libmikmod3 libsdl-mixer1.2 libsdl1.2debian
REMOVING:
blockattack
Summary:
Upgrading: 0, Installing: 6, Removing: 1, Not Upgrading: 73
Download size: 8,078 kB
Space needed: 4,102 kB / 34.2 GB available
Continue? [Y/n] y
Get:1 http://deb.debian.org/debian trixie/main amd64 libmikmod3 amd64 3.3.13-1 [135 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libsdl1.2debian amd64 1.2.68-3 [85.9 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 libsdl-mixer1.2 amd64 1.2.12-19+b1 [81.6 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 blockout2 amd64 2.5+dfsg1-2 [4,228 kB]
Get:5 http://deb.debian.org/debian trixie/main amd64 cuyo-data all 2.1.0-1 [3,372 kB]
Get:6 http://deb.debian.org/debian trixie/main amd64 cuyo amd64 2.1.0-1 [175 kB]
Fetched 8,078 kB in 1s (12.5 MB/s)
(Reading database ... 166139 files and directories currently
installed.)
Removing blockattack (2.9.0-1+b1) ...
Selecting previously unselected package libmikmod3:amd64.
(Reading database ... 166112 files and directories currently

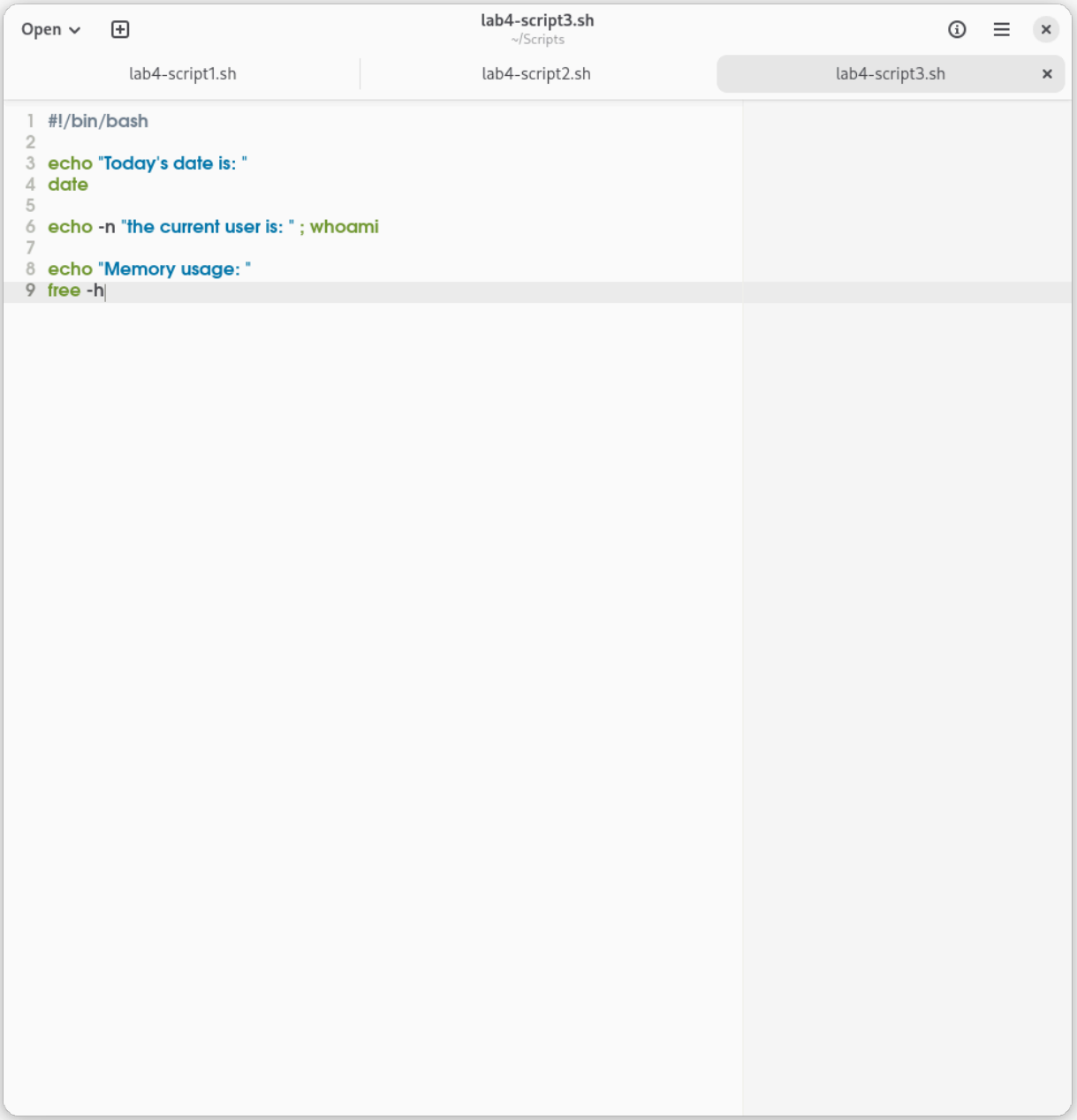

1 / 1
Tilix: d13@cis106desktop: ~
1: d13@cis106desktop: ~
d13@cis106desktop:~$ sudo apt remove cuyo
The following packages were automatically installed and are no longer required:
cuyo-data libopusfile0 libsdl2-image-2.0-0 libsdl2-ttf-2.0-0
libfmt10 libphysfs1 libsdl2-mixer-2.0-0 libxmp4
Use 'sudo apt autoremove' to remove them.
REMOVING:
cuyo
Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 73
Freed space: 489 kB
Continue? [Y/n] y
(Reading database ... 167209 files and directories currently installed.)
Removing cuyo (2.1.0-1) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
d13@cis106desktop:~$ sudo apt purge cuyo
The following packages were automatically installed and are no longer required:
cuyo-data libopusfile0 libsdl2-image-2.0-0 libsdl2-ttf-2.0-0
libfmt10 libphysfs1 libsdl2-mixer-2.0-0 libxmp4
Use 'sudo apt autoremove' to remove them.
REMOVING:
cuyo*
Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 73
Space needed: 0 B / 34.2 GB available
Continue? [Y/n] y
(Reading database ... 167201 files and directories currently installed.)

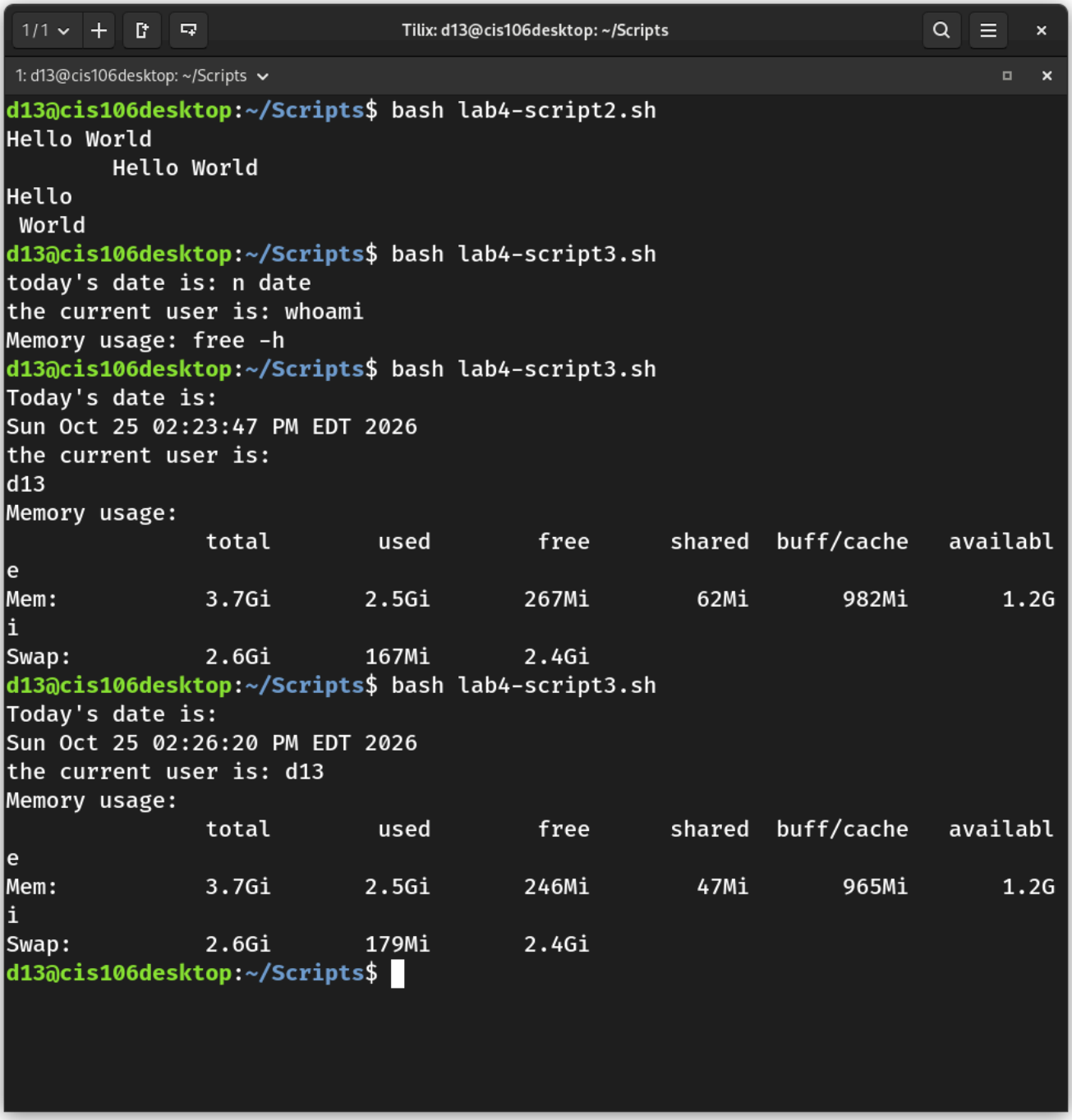

1 / 1
Tilix: d13@cis106desktop: ~
1: d13@cis106desktop: ~
d13@cis106desktop:~$ sudo apt clean && sudo apt autoclean && sudo apt autoremov
e && sudo apt autopurge
REMOVING:
cuyo-data libopusfile0 libsdl2-image-2.0-0 libsdl2-ttf-2.0-0
libfmt10 libphysfs1 libsdl2-mixer-2.0-0 libxmp4
Summary:
Upgrading: 0, Installing: 0, Removing: 8, Not Upgrading: 73
Freed space: 7,496 kB
Continue? [Y/n] y
(Reading database ... 167201 files and directories currently installed.)
Removing cuyo-data (2.1.0-1) ...
Removing libfmt10:amd64 (10.1.1+ds1-4) ...
Removing libsdl2-mixer-2.0-0:amd64 (2.8.1+dfsg-2) ...
Removing libopusfile0:amd64 (0.12-4+b3) ...
Removing libphysfs1:amd64 (3.0.2-6+b2) ...
Removing libsdl2-image-2.0-0:amd64 (2.8.8+dfsg-1+deb13u1) ...
Removing libsdl2-ttf-2.0-0:amd64 (2.24.0+dfsg-2) ...
Removing libxmp4:amd64 (4.6.2-2) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u4) ...
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 73
d13@cis106desktop:~$

Question 2

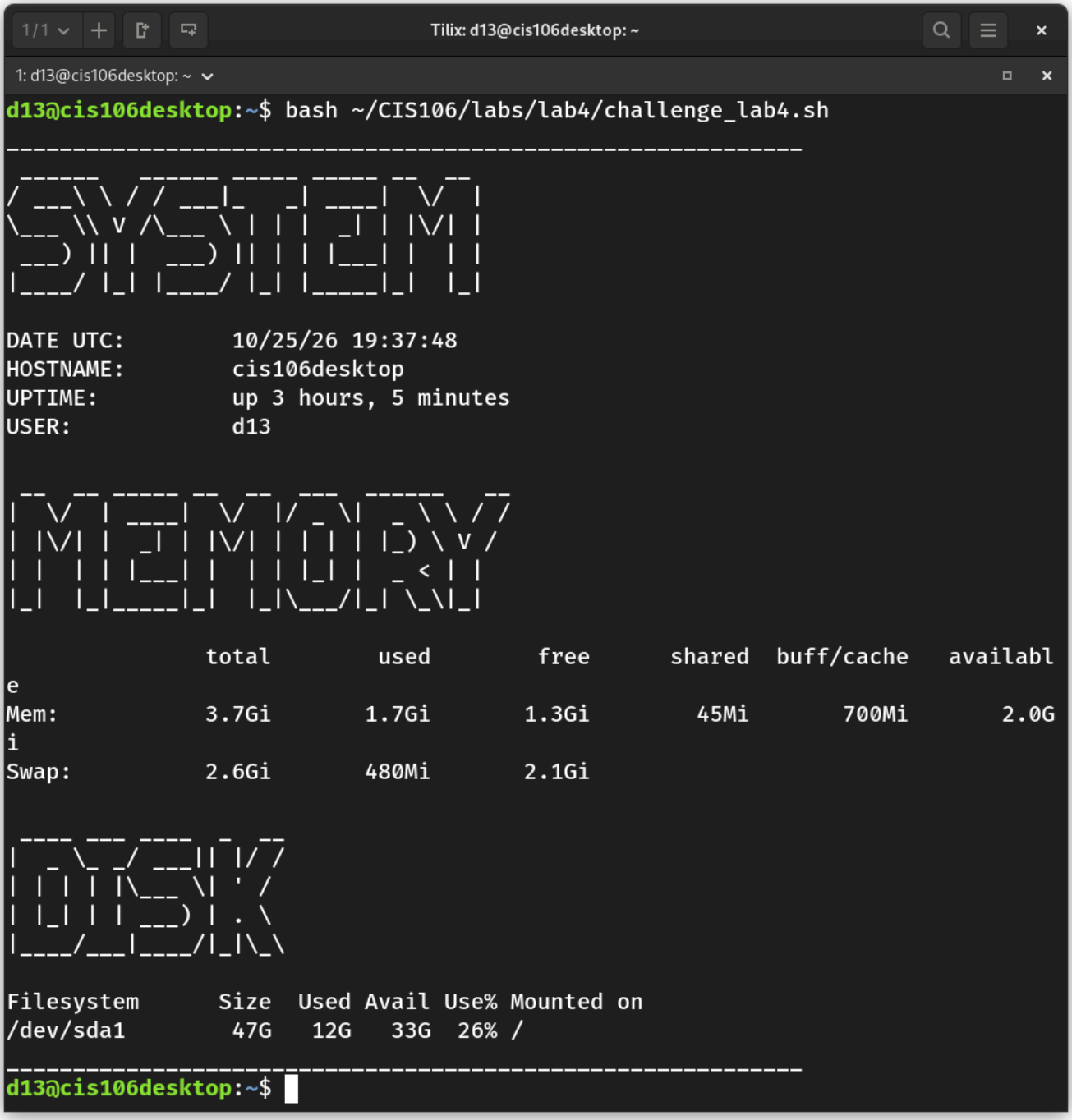

Open
lab4-script1.sh
~/Scripts
lab4-script1.sh
lab4-script2.sh
lab4-script3.sh
1 #!/bin/bash
2
3


Open
lab4-script2.sh
~/Scripts
lab4-script1.sh
lab4-script2.sh
lab4-script3.sh
#!/bin/bash
echo Hello World
echo -e "\tHello World"
echo -e "Hello\n World"


Open
lab4-script3.sh
~/Scripts
lab4-script1.sh
lab4-script2.sh
lab4-script3.sh
#!/bin/bash
echo "Today's date is: "
date
echo -n "the current user is: " ; whoami
echo "Memory usage: "
free -h


1 / 1
Tilix: d13@cis106desktop: ~/Scripts
1: d13@cis106desktop: ~/Scripts
d13@cis106desktop:~/Scripts$ bash lab4-script2.sh
Hello World
Hello World
Hello
World
d13@cis106desktop:~/Scripts$ bash lab4-script3.sh
today's date is: n date
the current user is: whoami
Memory usage: free -h
d13@cis106desktop:~/Scripts$ bash lab4-script3.sh
Today's date is:
Sun Oct 25 02:23:47 PM EDT 2026
the current user is:
d13
Memory usage:
total used free shared buff/cache availabl
e
Mem: 3.7Gi 2.5Gi 267Mi 62Mi 982Mi 1.2G
i
Swap: 2.6Gi 167Mi 2.4Gi
d13@cis106desktop:~/Scripts$ bash lab4-script3.sh
Today's date is:
Sun Oct 25 02:26:20 PM EDT 2026
the current user is: d13
Memory usage:
total used free shared buff/cache availabl
e
Mem: 3.7Gi 2.5Gi 246Mi 47Mi 965Mi 1.2G
i
Swap: 2.6Gi 179Mi 2.4Gi
d13@cis106desktop:~/Scripts$

Question 3


1 / 1
Tilix: d13@cis106desktop: ~
1: d13@cis106desktop: ~
d13@cis106desktop:~$ bash ~/CIS106/labs/lab4/challenge_lab4.sh
DATE UTC: 10/25/26 19:37:48
HOSTNAME: cis106desktop
UPTIME: up 3 hours, 5 minutes
USER: d13
total used free shared buff/cache availabl
e
Mem: 3.7Gi 1.7Gi 1.3Gi 45Mi 700Mi 2.0G
i
Swap: 2.6Gi 480Mi 2.1Gi
Filesystem Size Used Avail Use% Mounted on
/dev/sda1 47G 12G 33G 26% /
d13@cis106desktop:~$

Open

challenge_lab4.sh
~/CIS106/labs/lab4

```
#!/bin/bash

echo "_____________________________________________________"
figlet "SYSTEM"

echo "DATE UTC:       $(date -u '+%m/%d/%y %H:%M:%S')"
echo "HOSTNAME:       $(hostname)"
echo "UPTIME:         $(uptime -p)"
echo "USER:           $(whoami)"

echo ""
figlet "MEMORY"
free -h

echo ""
figlet "DISK"
df -h /

echo "_____________________________________________________"

```